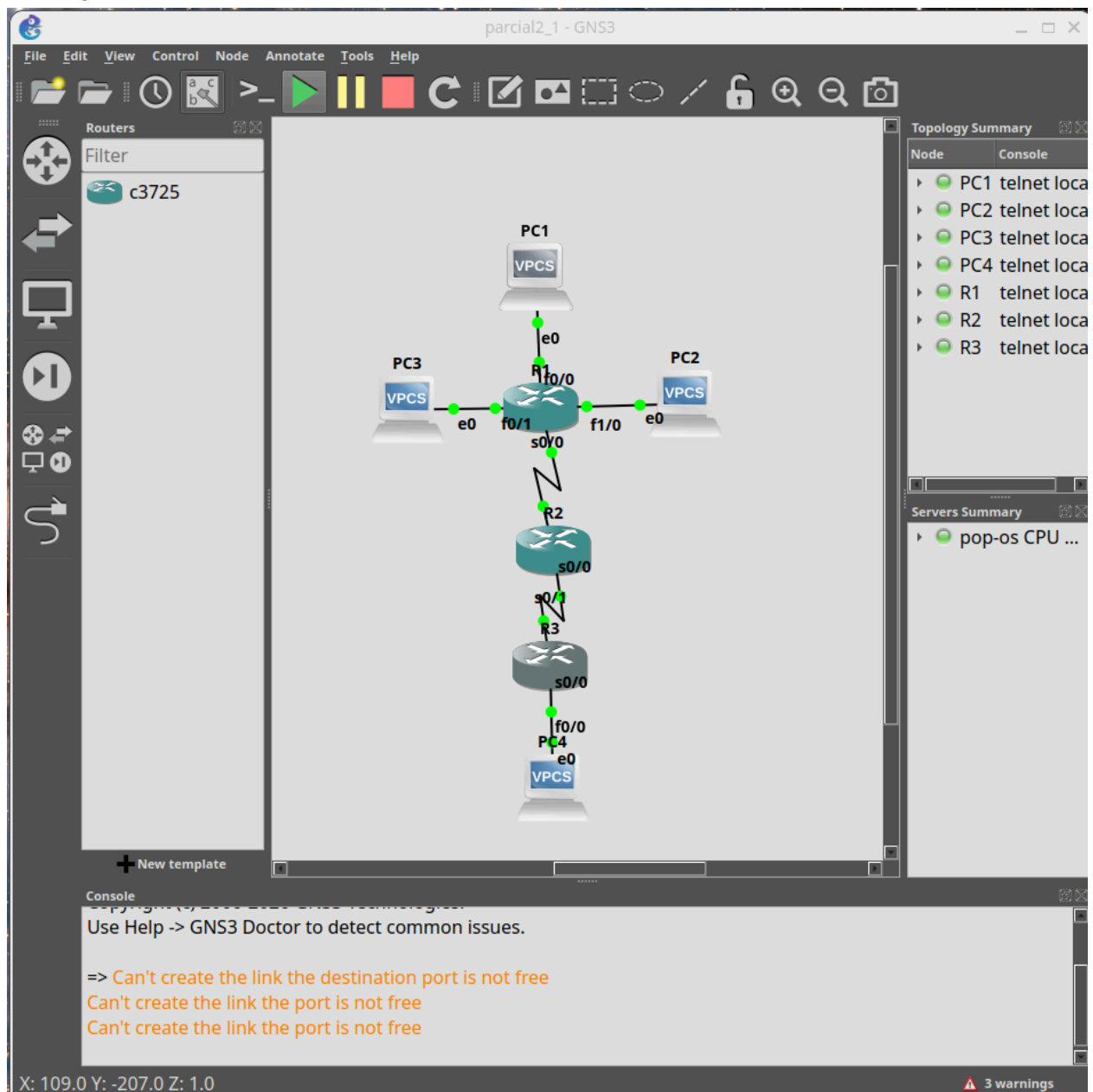


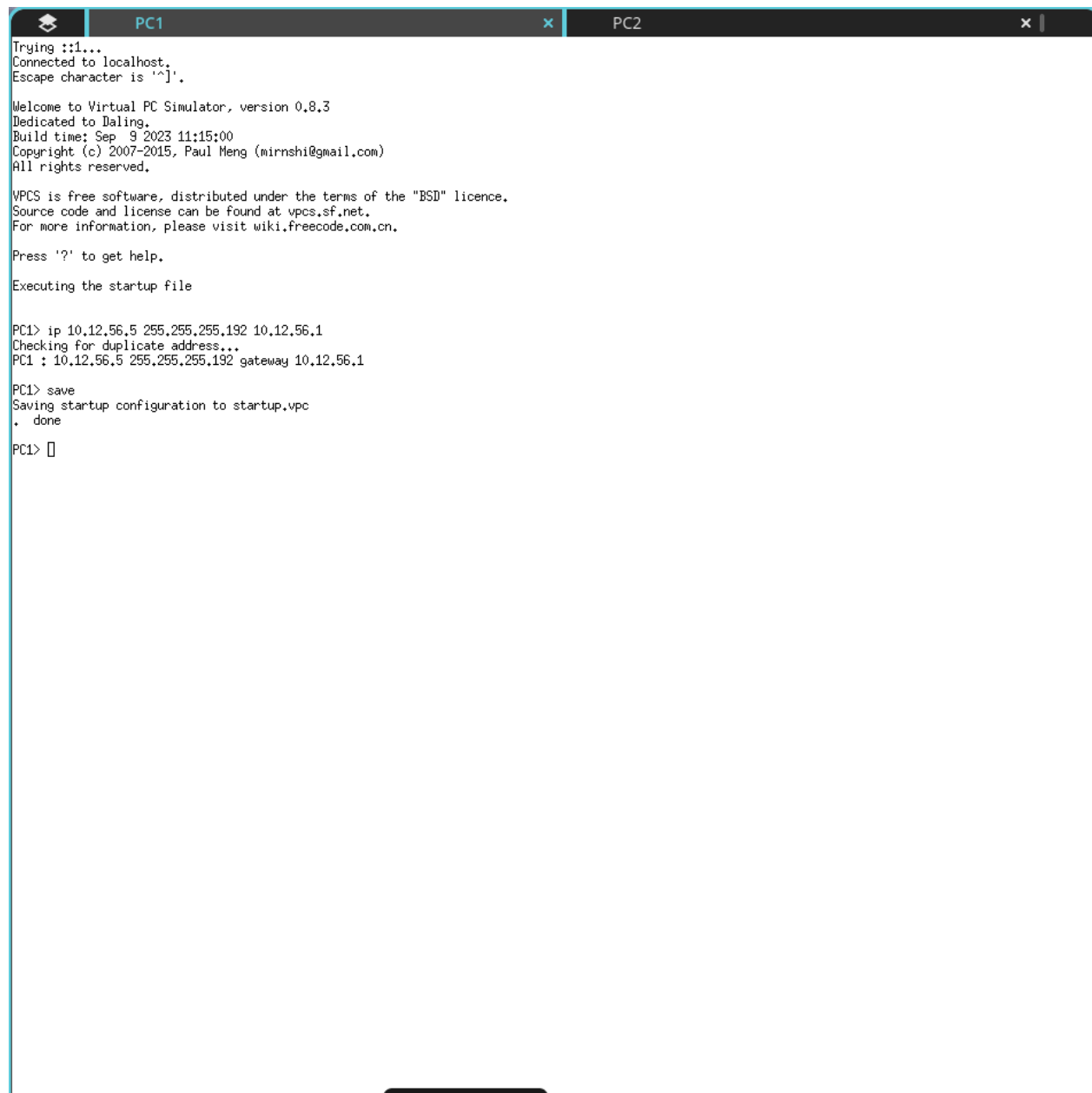
Ejercicio 1\

Topologia



PC

```

A screenshot of a Virtual PC Simulator window. The window has a dark title bar with two tabs: 'PC1' and 'PC2'. The 'PC1' tab is active. The main area is a black terminal window with white text. The text shows the simulator's startup sequence, including connection status, version information, and copyright details. It then shows the execution of a startup file, which sets the IP address to 10.12.56.5 and the gateway to 10.12.56.1. Finally, the user enters the 'save' command, and the simulator saves the configuration to a file named 'startup.vpc'. The prompt 'PC1>' is visible at the bottom of the terminal.
PC1
Trying ::1...
Connected to localhost.
Escape character is '^]'.

Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC1> ip 10.12.56.5 255.255.255.192 10.12.56.1
Checking for duplicate address...
PC1 : 10.12.56.5 255.255.255.192 gateway 10.12.56.1

PC1> save
Saving startup configuration to startup.vpc
. done

PC1> 
```

```
PC1 x PC2 x
Trying ::1...
Connected to localhost.
Escape character is '^]'.

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Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
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All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC2> ip 10.12.56.70 255.255.255.224 10.12.56.65
Checking for duplicate address...
PC2 : 10.12.56.70 255.255.255.224 gateway 10.12.56.65

PC2> save
Saving startup configuration to startup.vpc
. done

PC2> █
```

```

PC1
PC2
PC3
Trying ::1...
Connected to localhost.
Escape character is '^]'.

Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC3> 10.12.56.114 255.255.255.240 10.12.56.113
Bad command: "10.12.56.114 255.255.255.240 10.12.56.113". Use ? for help.

PC3> ip 10.12.56.114 255.255.255.240 10.12.56.113
Checking for duplicate address...
PC3 : 10.12.56.114 255.255.255.240 gateway 10.12.56.113

PC3> save
Saving startup configuration to startup.vpc
. done

PC3> 
```

```

PC1 x PC2 x PC3 x PC4 x
Trying ::1...
Connected to localhost.
Escape character is '^]'.

Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep 9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC4> ip 10.12.56.114 255.255.255.248 10.12.56.113
Checking for duplicate address...
PC4 : 10.12.56.114 255.255.255.248 gateway 10.12.56.113

PC4> save
Saving startup configuration to startup.vpc
. done

PC4> 
```

Routes

```

R1
*Mar 1 00:00:03.955: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to down
*Mar 1 00:00:03.971: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
*Mar 1 00:00:03.995: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to down
*Mar 1 00:00:03.995: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1, changed state to down
*Mar 1 00:00:03.999: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to down
R1#
R1#
R1#
R1#enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 10.12.56.1
% Incomplete command.

R1(config-if)#ip address 10.12.56.1 255.255.255.192
R1(config-if)#no shutdown
R1(config-if)#exit
*Mar 1 00:02:01.483: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:02:02.483: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#exit
R1(config)#interface FastEthernet1/0
R1(config-if)#ip address 10.12.56.65 255.255.255.224
R1(config-if)#no shutdown
R1(config-if)#exit
*Mar 1 00:02:55.739: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 1 00:02:56.739: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R1(config-if)#exit
R1(config)#interface FastEthernet0/1
R1(config-if)#ip address 10.12.56.97 255.255.255.224
R1(config-if)#no shutdown
R1(config-if)#exit
*Mar 1 00:03:48.651: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:03:49.651: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R1(config-if)#exit
R1(config)#interface Serial0/0
R1(config-if)#ip address 10.12.56.121 255.255.255.252
% 10.12.56.120 overlaps with FastEthernet0/1
R1(config-if)#no shutdown
% 10.12.56.120 overlaps with FastEthernet0/1
Serial0/0: incorrect IP address assignment
R1(config-if)#exit
R1(config)#end
R1#
*Mar 1 00:04:56.667: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 10.12.56.1      YES manual up          up
Serial0/0       10.12.56.121    YES manual administratively down down
FastEthernet0/1 10.12.56.97     YES manual up          up
Serial0/1       unassigned      YES unset  administratively down down
FastEthernet1/0 10.12.56.65     YES manual up          up
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip route 10.12.56.112 255.255.255.248 10.12.56.122
R1(config)#ip route 10.12.56.124 255.255.255.252 10.12.56.122
R1(config)#write memory
^
% Invalid input detected at '^' marker.

R1(config)#wr
% Incomplete command.

R1(config)#end
R1#wr
*Mar 1 00:07:47.955: %SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
R1#

```

```

Cisco IOS Software, 3700 Software (C3725-ADVENTERPRISEK9-M), Version 12.4(25d),
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2010 by Cisco Systems, Inc.
Compiled Wed 18-Aug-10 07:55 by prod_rel_team
*Mar 1 00:00:02.011: %SNMP-5-COLDSTART: SNMP agent on host R2 is undergoing a c
*Mar 1 00:00:02.171: %LINEPROTO-5-UPDOWN: Line protocol on Interface IPv6-mpls,
*Mar 1 00:00:02.951: %LINK-5-CHANGED: Interface FastEthernet0/0, changed state
*Mar 1 00:00:02.975: %LINK-5-CHANGED: Interface FastEthernet0/1, changed state
*Mar 1 00:00:02.999: %LINK-5-CHANGED: Interface Serial0/0, changed state to adm
*Mar 1 00:00:02.999: %LINK-5-CHANGED: Interface Serial0/1, changed state to adm
*Mar 1 00:00:03.951: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet
*Mar 1 00:00:03.975: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet
*Mar 1 00:00:03.999: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0,
*Mar 1 00:00:03.999: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1,
R2#
R2#
R2#
R2#
R2#
R2#enable
R2#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#interface Serial0/0
R2(config-if)#ip address 10.12.56.122 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#
*Mar 1 00:05:44.215: %LINK-3-UPDOWN: Interface Serial0/0, changed state to up
R2(config-if)#
*Mar 1 00:05:45.219: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up
R2(config-if)#exit
R2(config)#interface Serial0/1
R2(config-if)#ip address 10.12.56.125 255.255.255.25
*Mar 1 00:06:11.983: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to down
R2(config-if)#ip address 10.12.56.125 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
*Mar 1 00:06:24.919: %LINK-3-UPDOWN: Interface Serial0/1, changed state to up
R2(config-if)#exit
*Mar 1 00:06:25.923: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1, changed state to up
R2(config-if)#exit
R2(config)#ip route 10.12.56.0 255.255.255.192 10.12.56.121
*Mar 1 00:06:52.003: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1, changed state to down
R2(config)#ip route 10.12.56.0 255.255.255.192 10.12.56.121
R2(config)#ip route 10.12.56.64 255.255.255.224 10.12.56.121
R2(config)#ip route 10.12.56.96 255.255.255.240 10.12.56.121
R2(config)#ip route 10.12.56.112 255.255.255.248 10.12.56.126
R2(config)#end
R2#ur
*Mar 1 00:09:12.895: %SYS-5-CONFIG_I: Configured from console by console
R2#ur
Building configuration...
[OK]
R2#show ip route
```

```
R1      x      R2      x      R3      x
*Mar 1 00:00:01.787: %SW_VLAN-4-IFS_FAILURE: VLAN manager encountered file op
eration error: call = ifs_open/read / code = 3588 (No device available)
/ bytes transferred = 0
*Mar 1 00:00:01.851: %SYS-5-CONFIG_I: Configured from memory by console
*Mar 1 00:00:01.915: %LINEPROTO-5-UPDOWN: Line protocol on Interface VoIP-Nul
l0, changed state to up
*Mar 1 00:00:01.991: %SYS-5-RESTART: System restarted --
Cisco IOS Software, 3700 Software (C3725-ADVENTERPRISEK9-M), Version 12.4(25d)
, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2010 by Cisco Systems, Inc.
Compiled Wed 18-Aug-10 07:55 by prod_rel_team
*Mar 1 00:00:01.999: %SNMP-5-COLDSTART: SNMP agent on host R3 is undergoing a
cold start
*Mar 1 00:00:02.163: %LINEPROTO-5-UPDOWN: Line protocol on Interface IPv6-mp1
s, changed state to up
*Mar 1 00:00:02.943: %LINK-5-CHANGED: Interface FastEthernet0/0, changed stat
e to administratively down
*Mar 1 00:00:02.967: %LINK-5-CHANGED: Interface FastEthernet0/1, changed stat
e to administratively down
*Mar 1 00:00:02.987: %LINK-5-CHANGED: Interface Serial0/0, changed state to a
dministratively down
*Mar 1 00:00:02.987: %LINK-5-CHANGED: Interface Serial0/1, changed state to a
dministratively down
*Mar 1 00:00:03.943: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthe
rnet0/0, changed state to down
*Mar 1 00:00:03.967: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthe
rnet0/1, changed state to down
*Mar 1 00:00:03.987: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/
0, changed state to down
*Mar 1 00:00:03.987: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/
1, changed state to down
R3#
R3#
R3#
R3#
R3#
R3#
R3#
R3#
R3#enable
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#interface FastEthernet0/0
R3(config-if)#ip address 10.12.56.113 255.255.255.248
R3(config-if)#no shutdown
R3(config-if)#exit
*Mar 1 00:08:02.423: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:08:03.423: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R3(config-if)#exit
R3(config)#interface Serial0/0
R3(config-if)#ip address 10.12.56.126 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
*Mar 1 00:08:47.179: %LINK-3-UPDOWN: Interface Serial0/0, changed state to up
R3(config-if)#exit
*Mar 1 00:08:48.183: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up
R3(config-if)#exit
R3(config)#ip route 10.12.56.0 255.255.255.192 10.12.56.125
R3(config)#ip route 10.12.56.64 255.255.255.224 10.12.56.125
R3(config)#ip route 10.12.56.96 255.255.255.240 10.12.56.125
R3(config)#ip route 10.12.56.120 255.255.255.252 10.12.56.125
R3(config)#end
R3#
*Mar 1 00:10:38.263: %SYS-5-CONFIG_I: Configured from console by console
R3#
Building configuration...
[OK]
R3#
```

VLSM				
	Red	Maroon	Infra	Broadcast
10.12.56.0				
10.12.56.0	10.12.56.0	255.255.255.192	10.12.56.1 - 10.12.56.62	10.12.56.63
10.12.56.64	10.12.56.64	255.255.255.224	10.12.56.65 - 10.12.56.94	10.12.56.95
10.12.56.96	10.12.56.96	255.255.255.240	10.12.56.97 - 10.12.56.110	10.12.56.111
10.12.56.112	10.12.56.112	255.255.255.248	10.12.56.113 - 10.12.56.128	10.12.56.129
10.12.56.129	10.12.56.129	255.255.255.252	10.12.56.131 - 10.12.56.172	10.12.56.173
10.12.56.173	10.12.56.173	255.255.255.252	10.12.56.175 - 10.12.56.126	10.12.56.127

Ping

```
Trying ::1...
Connected to localhost.
Escape character is '^]'.
ping 10.12.56.114
Cannot resolve 10.12.56.114ping

PC1> trace 10.12.56.114
Trace to 10.12.56.114, 8 hops max, press Ctrl+C to stop
 1  10.12.56.1    7.488 ms  10.090 ms  9.459 ms
 2  10.12.56.122  9.922 ms  9.300 ms  9.822 ms
 3  10.12.56.126  9.022 ms  9.626 ms  10.010 ms
 4  *10.12.56.114 19.674 ms (ICMP type:3, code:3, Destination port unreachable)

PC1> ping 10.12.56.114

84 bytes from 10.12.56.114 icmp_seq=1 ttl=61 time=16.927 ms
84 bytes from 10.12.56.114 icmp_seq=2 ttl=61 time=19.612 ms
84 bytes from 10.12.56.114 icmp_seq=3 ttl=61 time=18.190 ms
84 bytes from 10.12.56.114 icmp_seq=4 ttl=61 time=19.737 ms
84 bytes from 10.12.56.114 icmp_seq=5 ttl=61 time=9.403 ms

PC1> █
```

